

The deterministic chart posterior ratio and the corner-vanishing regime

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Abstract

Astra #14 rank 2. For a fixed phase x and amplitudes y_φ, y_1 with $y_{1,00} > 0$ and equal starting exponents p : if $y_{\varphi,00} \neq 0$ then $Z[\varphi](N)/Z[1](N) \rightarrow y_{\varphi,00}/y_{1,00}$ (`chart_posterior_tendsto_const`); if $y_{\varphi,00} = 0$ but the constant coefficient $B_p^\varphi \neq 0$ then $(Z[\varphi]/Z[1]) \log N \rightarrow B_p^\varphi/A_p^1$ (`chart_posterior_tendsto_log`). The log coefficient vanishes exactly when the amplitude vanishes at the corner (`leading_log_coeff_eq_zero_iff`). So the posterior expectation of an observable vanishing at the corner decays like $1/\log N$, not like a power of N : Astra #14's correction of the naive vanishing-order heuristic, now a theorem. Zero sorry/axiom.

1 The things formalised

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theorem leading_log_coeff_eq_zero_iff ... :
  canoA h h k k (fun i j s => ampCoeff x y (i, j) s) p = 0 y (0, 0) = 0
theorem chart_posterior_tendsto_const ... (hy 0 : 0 < y (0,0)) (hy 0 : y (0,0) 0) :
  Tendsto (fun N => twoDamp ... (anaAmp (ampCoeff x y) b)
    / twoDamp ... (anaAmp (ampCoeff x y) b))
    atTop ( (y (0,0) / y (0,0)))
theorem chart_posterior_tendsto_log ... (hy 0 : y (0,0) = 0)
  (hB : canonB ... (ampCoeff x y) ... p 0) :
  Tendsto (fun N => twoDamp ... y ... / twoDamp ... y ... * Real.log N) atTop
  ( (canonB ... y ... p / canoA ... y ... p))
```

2 Mathematics

By unit 167, $Z[\varphi] \sim A_p^\varphi N^{-p} \log N$ if $A_p^\varphi \neq 0$ and $Z[\varphi] \sim B_p^\varphi N^{-p}$ if $A_p^\varphi = 0 \neq B_p^\varphi$, while $Z[1] \sim A_p^1 N^{-p} \log N$ with $A_p^1 > 0$. Since $A_p = \frac{y_{00}}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s x_{00}} ds$ with a positive integral, $A_p = 0$ iff $y_{00} = 0$, and the ratio A_p^φ/A_p^1 is $y_{\varphi,00}/y_{1,00}$. The quotient calculus of unit 166 then gives the two limits.

3 Paper-facing meaning

`cor:empirical_expectation`'s leading term $A_{i_0,j_0}(\varphi)/B_{k_0,p_0}$ with different leading indices: at chart level the first case beyond the generic one is an observable vanishing at the corner, and the theorem shows the leading indices differ in the log power, not in the exponent of N — the posterior expectation decays like $1/\log N$. This corrects the tempting reading “vanishing to order m at the

singularity improves the power of N ” (Astra #14: the power only improves under coordinatewise divisibility, which raises the exponents $(h_i + 1)/k_i$ to $(h_i + a_i + 1)/k_i$ and needs the face terms). The stochastic version (random phase converging in distribution) follows the same pattern and is left for a later unit; the formulas for B_p^φ (the face/finite-part coefficients of unit 121) remain implicit here.

4 Lean notes

- Naming a local hypothesis `h` shadows the exponent variable `h` : and breaks later applications (“expected ”); use distinct names (`hE`).
- `tendsto_quotient_mul_log_of_eq` with $r = 0$ takes the denominator equivalence at $r + 1$, which is definitionally 1.
- The coefficient-ratio identity is `field_simp` after `set M := logMoment ....`